Matteo **Todeschi**

Master's student · Tech Enthusiast

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Summary

Computer Science & Engineering graduate from the University of Bologna, currently pursuing a Master's Degree at Cesena Campus. Experienced in distributed microservice architectures and real-time systems through applied research for the Horizon Europe eBRT2030 initiative. Driven by solving complex challenges, developing scalable and maintainable systems, and thriving in collaborative academic and engineering environments.

Skills

Languages	Italian (Native), English (Professional Working Proficiency, CEFR B2)
Programming	Java, JavaScript, Python, PHP, SQL, Kotlin, C, C++
Frameworks & Tools	Docker, FastAPI, Vue.js, TypeScript, Git, Linux, Node.js, Traefik, Keycloak, MQTT

Education

University of Bologna (Alma Mater Studiorum)

Cesena, Italy

Master of Science in Computer Science and Engineering — *Sept. 2026 – Present*

University of Bologna (Alma Mater Studiorum)

Cesena, Italy

Bachelor of Science in Computer Science and Engineering — Final Grade: 109/110 — *Sept. 2023 – Oct. 2026*

- Thesis: "Design and Development of a Microservice Architecture and Public Transport Digital Twin: The Rimini Metromare Case Study".
- Advisor: Dr. Giovanni Delnevo — Co-Advisors: Dr. Manuel Andruccioli, Dr. Kelvin Olaiya.

Liceo Scientifico Statale "Alfredo Oriani"

Ravenna, Italy

High School Diploma: Applied Sciences — Final Grade: 98/100 — *Sept. 2018 – July 2023*

Experience

University of Bologna & Start Romagna

Cesena / Rimini, Italy

Software Engineering Intern & Graduate Researcher — EU Project eBRT2030 — *Oct. 2025 – Oct. 2026*

- Engineered a distributed microservice architecture and event-driven Digital Twin (MQTT, FastAPI, Redis, MongoDB) under the Horizon Europe eBRT2030 project to bridge sparse AVM telemetry with real-time kinematic vehicle interpolation.
- Developed a real-time operator monitoring dashboard using Vue 3.5, TypeScript, Pinia, and WebSockets, integrating topological SVG and Leaflet map tracking.
- Containerized an isolated ML service for passenger demand forecasting and automated anomaly detection, validated through SUMO simulations.

Technical Projects

QuizLab — Zero-Trust Quiz & Flashcard Platform

quizlab

Docker, Traefik, Keycloak, Envoy, Node.js — *2026*

- Architected an Identity-Aware Zero-Trust platform isolating microservices via Traefik Edge Gateway, OAuth2-Proxy, and Keycloak OIDC.
- Implemented an intra-service encrypted Service Mesh with local Envoy mTLS sidecars across isolated Docker networks.

Moovie — Android Movie Discovery & Social App

moovie

Kotlin, Jetpack Compose, Room SQLite, Supabase, Ktor — *2026*

- Developed a modern Android app featuring an offline-first architecture using Room SQLite, Koin DI, and Ktor for TMDB APIs.
- Implemented cloud sync via Supabase, reactive state with Coroutines/StateFlow, biometric auth, and Google Maps SDK.

Graphly — Interactive Graph Algorithm Visualizer

graphly

JavaScript (ES6+), SVG, HTML5 Canvas, CSS3 — *2026*

- Developed a high-performance tool to visualize graph algorithms (Dijkstra, Kruskal, Ford-Fulkerson) step-by-step.
- Implemented core logic without external frameworks to ensure deep understanding of DOM manipulation and state.

JavaDyno — Engine Dynamometer & Telemetry Suite

java-dyno

Java, JavaFX/Swing, Multithreading, MVC Architecture — *2024*

- Co-developed in an Agile team of 4, using concurrent threads and latches to decouple simulation from UI rendering.
- Built a modular JavaFX UI integrating dynamic JFreeChart telemetry, Medusa gauges, and a central Alert Monitor.

Smart Tank — Distributed IoT Monitoring & Control System

smart-tank-monitoring-system

ESP32, Arduino Uno, MQTT, Serial, Node.js, WebSockets — *2025*

- Architected a distributed IoT system bridging an ESP32 telemetry node (ultrasonic level sensing via MQTT) and an Arduino Uno valve actuator.
- Engineered a Node.js central unit bridging MQTT with bidirectional Serial communication, real-time WebSockets, and FSM-driven soft takeover.